

Statistical Arbitrage

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Problem 1: The OU Process

1.

$$\begin{aligned}dX_t &= \theta(\mu - X_t)dt + \sigma dW_t, \quad \theta, \sigma > 0 \\d(e^{\theta t} X_t) &= e^{\theta t} dX_t + \theta e^{\theta t} X_t dt \\d(e^{\theta t} X_t) &= e^{\theta t} (\theta(\mu - X_t)dt + \sigma dW_t) + \theta e^{\theta t} X_t dt \\d(e^{\theta t} X_t) &= \theta e^{\theta t} \mu dt + \sigma e^{\theta t} dW_t \\e^{\theta t} X_t - X_0 &= \mu(e^{\theta t} - 1) + \sigma \int_0^t e^{\theta s} dW_s \\X_t &= \mu - (X_0 - \mu)e^{-\theta t} + \sigma \int_0^t e^{-\theta(t-s)} dW_s\end{aligned}$$

2. We begin by investigating the stochastic term in the previous result:

$$\begin{aligned}X_t &= \alpha + \beta, \quad \alpha = \mu - (X_0 - \mu)e^{-\theta t}, \quad \beta = \sigma \int_0^t e^{-\theta(t-s)} dW_s \\&\sim N\left(0, \sigma^2 \int_0^t e^{-2\theta(t-s)} ds\right) \\&\sim N\left(0, \frac{\sigma^2}{2\theta} [1 - e^{-2\theta t}]\right)\end{aligned}$$

$$\begin{aligned}E(X_t) &= E\left(\mu - (X_0 - \mu)e^{-\theta t} + \sigma \int_0^t e^{-\theta(t-s)} dW_s\right) \\&= \mu - (X_0 - \mu)e^{-\theta t} + E\left(\sigma \int_0^t e^{-\theta(t-s)} dW_s\right) \\&= \mu - (X_0 - \mu)e^{-\theta t}\end{aligned}$$

$$\begin{aligned}X_t \text{Var}(X_t) &= E(X_t^2) - E(X_t)^2 \\&= E(\alpha^2 + 2\alpha\beta + \beta^2) - \alpha^2 \\&= E(\beta^2) \\&= \frac{\sigma^2}{2\theta} [1 - e^{-2\theta t}]\end{aligned}$$

This implies that the stationary distribution for $\lim_{t \rightarrow \infty} X_t$ is

$$\lim_{t \rightarrow \infty} X_t \sim N\left(\mu, \frac{\sigma^2}{2\theta}\right)$$

3. Half-life of a deviation. First we define a deviation as a perturbation from an initial condition. Since this is a stochastic timeseries, we can inject this deviation at any point in the timeseries. We define its half-life λ as how long it takes for that deviation to decrease to half of its original value.

$$X_t|X_0 + d = \mu + (X_0 + d - \mu)e^{\theta t} + \sigma \int_0^t e^{-\theta(t-s)} dW_s$$

$$X_t|X_0 + d = (X_t|X_0) + de^{-\theta t}$$

We find that the deviation follows an exponential decay curve given by $e^{-\theta t}$. The half life of this is easy to measure, we have

$$\frac{1}{2} = e^{-\theta \lambda}$$

$$\lambda = \frac{\ln(2)}{\theta}$$

4. Exact Discretization: We find the exact discretization by choosing $t = \Delta t$ in the original solution to the SDE. We find

$$X_{n+1}|X_n = \mu + (X_n - \mu)e^{-\Delta t \theta} + \int_0^{\Delta t} e^{-\theta(\Delta t-s)} dW_s$$

$$= \mu(1 - e^{-\Delta t \theta}) + X_n e^{-\Delta t \theta} + \int_0^{\Delta t} e^{-\theta(\Delta t-s)} dW_s$$

$$= a + bX_n + \epsilon_k$$

Here we have shown that $a = \mu(1 - b)$, $b = e^{-\Delta t \theta}$ and $\epsilon_k = \int_0^{\Delta t} e^{-\theta(\Delta t-s)} dW_s$. We have already shown (using Δt to replace t) that $\epsilon_k \sim N\left(0, \frac{\sigma^2}{2\theta}(1 - b^2)\right)$. We invert these equations to find,

$$\theta = -\frac{\ln(b)}{\Delta t}$$

$$\mu = \frac{a}{1 - b}$$

$$\sigma = -\frac{2\text{Var}(\epsilon_k) \ln(b)}{\Delta t(1 - b^2)}$$

Problem 2: Simulation and Estimation

1. Simulating Euler-Maruyama
2. Derive MLE of (θ, μ, σ) . Show it is equivalent to the least squares regression of X_{k+1} on X_k .
3. Analyze the effectiveness of θ .